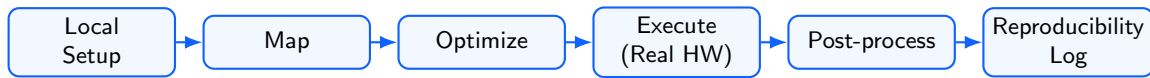


Week 6: First Qiskit Program on Real IBM Quantum Hardware

Bell State Experiment — Simulation to Real QPU + Reproducibility Logging

Course URL: <https://quantum.cloud.ibm.com/learning/en/courses/quantum-computing-in-practice>



1. IBM Quantum Setup

- Free account at quantum.ibm.com
- API token generation (Dashboard → API Token → Copy)
- Authentication via `QiskitRuntimeService.save_account()`
- Backend selection: `service.least_busy(simulator=False)`
- Open Plan: free access to real QPUs

2. The Experiment: Bell State

- Bell state: $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- Circuit: H on qubit 0 → $\text{CNOT}(0, 1)$ → `measure_all`
- The math: $|00\rangle \xrightarrow{H \otimes I} \text{superposition} \xrightarrow{\text{CNOT}} \text{entanglement}$
- Expected: only $|00\rangle$ and $|11\rangle$ with equal probability

3. Running on Real Hardware

- Transpilation to ISA circuit (`optimization_level=3`)
- Use `SamplerV2` from `qiskit_ibm_runtime` (NOT local sampler)
- Key differences from simulator:
 - Gate errors (decoherence, crosstalk)
 - Readout errors
 - Queue wait times
 - Non-zero probability for $|01\rangle$ and $|10\rangle$ due to noise
- Fidelity metric: $P(|00\rangle) + P(|11\rangle)$ — should be $\sim 0.95+$ on modern hardware

4. Simulator vs Hardware Comparison

Metric	Simulator	Real Hardware
$P(00\rangle)$	~ 0.500	~ 0.48
$P(11\rangle)$	~ 0.500	~ 0.47
$P(01\rangle)$	~ 0.000	$\sim 0.02 - 0.03$
$P(10\rangle)$	~ 0.000	$\sim 0.02 - 0.03$
Fidelity	~ 1.000	~ 0.95

5. Reproducibility Log

A reproducibility log documents everything needed to reproduce a quantum experiment:

- Table with categories: Experiment Info, Software Stack, Backend/Hardware, Circuit Details, Execution Parameters, Results Summary
- Why it matters: quantum hardware changes with each calibration cycle, noise varies, backends are updated
- Key items to record: Job ID, backend name, Qiskit version, calibration date, gate error rates, shots, optimization level

6. Qiskit Tools Used

- `QiskitRuntimeService` (IBM Quantum authentication)
- `SamplerV2` (real hardware execution)
- `BackendSamplerV2` (local simulator execution)
- `generate_preset_pass_manager` (transpilation)
- `AerSimulator` (baseline comparison)

Key Takeaways

1. **Real Hardware Has Noise:** Unlike simulators, real QPUs produce non-ideal results due to gate errors, decoherence, and readout noise.
2. **Transpilation is Essential:** Abstract circuits must be converted to ISA circuits matching the target hardware's basis gates and coupling map.
3. **Reproducibility Requires Documentation:** Quantum hardware properties change with calibration — always record backend, Job ID, software versions, and execution parameters.
4. **Bell State Validates Entanglement:** High correlation between $|00\rangle$ and $|11\rangle$ confirms quantum entanglement even on noisy hardware.
5. **The Qiskit Pattern is Universal:** Map → Optimize → Execute → Post-process works identically for simulation and real hardware; only the backend changes.

References

- IBM Quantum Computing in Practice course
- `lecture2.pdf` (course slides)